

Autopsy

Report on an Open-Source Digital Forensics Platform

1. Introduction

Autopsy is a digital forensics platform which is open source and has a graphic interface for The Sleuth Kit (TSK) and other forensic tools. It is put together to help investigators look at digital evidence and determine what transpired on a computer or any other digital source it supports. Autopsy is used in law enforcement, military, corporate and educational settings. Best to think of Autopsy as an exam and analysis platform rather than just a file recovery program. It provides a case management environment, evidence processing pipeline, forensic artifact analysis, searching, timeline analysis, visualization and report features. The base software is free and open source.

2. What Autopsy Actually Does

Autopsy takes in a forensic data source, run it through a set of analysis modules which it has, store the results of that analysis in a case database and then present that information to the investigator via a graphic interface. It works with disk images, local drives and sets of files. It supports image formats like raw/dd and E01. It also looks at common file systems, finds files, metadata, deleted content and application level artifacts.

Important functions include:

- Filesystem and disk-image analysis
- Deleted-file and unallocated-space analysis
- Keyword searching and indexing
- Hash-set analysis and filtering
- Web-browser artifact extraction
- Registry analysis
- Email analysis
- Timeline analysis
- Multimedia and EXIF analysis
- File-type identification and extension-mismatch detection
- Android artifact analysis
- Data carving
- Malware scanning
- Evidence tagging and bookmarking
- Report generation

3. Working Architecture

Autopsy is a modular platform. It has a graphical interface which is over The Sleuth Kit and other forensic tools. To get started data is added to a case which is run through one or more ingest modules. The modules do in depth analysis which they put into a central case database. Then the investigator looks over the results via the user interface and produces reports..

Layer / Component	Role
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Case layer	Creates and manages the investigation. A case can contain one or more data sources and has its own directory, database, configuration and generated reports.
Data-source layer	Accepts supported sources such as disk images, local drives and logical files.
The Sleuth Kit / filesystem layer	Provides low-level filesystem and disk-structure analysis and exposes files and filesystem metadata to higher-level processing.
Ingest framework	Runs analysis modules against the data source and its files. Modules can process hashes, keywords, web artifacts, recent activity, file types and many other artifact classes.
Central database	Stores structured analysis results and artifacts so that different modules and interface components can access and correlate them.
Analysis interface	Lets the examiner browse files, view artifacts, search, inspect metadata, create timelines, tag evidence and correlate findings.
Reporting layer	Collects selected results and produces reports such as HTML, Excel and body-file outputs; additional report modules can be added.

The central architectural idea is the **ingest-module pipeline**. Autopsy can run multiple modules in parallel, allowing different analysis tasks to be performed while the evidence is being processed. The results are then made available through the case interface. The platform is also extensible: developers can create ingest, content-viewer, result-viewer and report modules.

4. Working Procedure

Autopsy's standard workflow can be summarized as follows:

1. **Create a case:** create a container for the investigation.
2. **Add a data source:** add a forensic disk image, local drive, logical files or another supported source.
3. **Select ingest modules:** choose the analysis modules that should process the evidence.
4. **Run ingest:** Autopsy processes the evidence in the background and produces artifacts and alerts as results become available.
5. **Perform manual analysis:** inspect files, metadata, application artifacts, deleted files, keyword hits, hash hits and other results.
6. **Tag relevant evidence:** bookmark or tag important files and artifacts and add examiner comments where necessary.
7. **Correlate events:** use timelines, web artifacts, hashes, locations and other evidence to reconstruct activity.
8. **Generate a report:** export the selected findings into an appropriate report format.

This workflow separates automated processing from human interpretation. The software can locate and organize large amounts of evidence, but the examiner remains responsible for deciding which artifacts are significant and how they should be interpreted.

5. The Sleuth Kit and Autopsy

The Sleuth Kit is the underlying open-source collection of command-line libraries and tools used for low-level digital-forensic analysis, particularly filesystem and volume analysis. Autopsy provides a graphical, higher-level environment around TSK and other tools.

A simplified relationship is:

Forensic image → The Sleuth Kit / other forensic components → Autopsy ingest modules → central case database → graphical analysis → report.

This architecture allows the investigator to use low-level forensic capabilities without manually executing many separate command-line utilities. It also allows additional analysis modules to be integrated when a particular investigation requires them.

6. Ingest Modules

Ingest modules are one of the most important parts of Autopsy's architecture. They perform automated processing when a data source is added to a case.

Examples include:

- **Hash Lookup:** calculates or compares hashes and can identify known-good or known-bad files.
- **Keyword Search:** indexes content and searches for specified terms or patterns.
- **Web Artifacts:** extracts browser history, bookmarks and cookies.
- **Recent Activity:** identifies supported user-activity artifacts.
- **File Type Identification:** determines file types using content/signatures and detects extension mismatches.
- **Embedded File Extraction:** extracts files embedded inside supported containers.
- **Picture Analyzer:** processes image-related information.
- **PhotoRec Carver:** attempts file recovery from appropriate unallocated-space data.
- **Android Analyzer:** extracts supported Android artifacts.

Modules can be run in parallel, which allows Autopsy to begin returning useful results before every analysis task has completed.

7. Evidence and Artifact Analysis

Autopsy does not treat a forensic image as one large undifferentiated block. It identifies filesystem structures and files, then uses specialized modules to extract higher-level artifacts.

For example:

Disk image → filesystem → files → browser database → browser artifacts → timeline/search/correlation.

The examiner can also move in the opposite direction when validation is needed: from a parsed artifact back to the underlying file, database or filesystem object. This is important because automated artifact parsing should support, not replace, forensic examination.

8. Major Analysis Capabilities

Timeline analysis: Autopsy can present system events in a graphical timeline, allowing investigators to identify activity over time and correlate events.

Keyword search: Text can be extracted and indexed so investigators can search large evidence collections for names, phrases, URLs, identifiers and other case-relevant terms.

Hash analysis: Known-good files can be filtered out and known-bad files can be flagged using supported hash sets.

Web analysis: Browser history, bookmarks, cookies and other web artifacts can be extracted from supported browsers.

Registry analysis: Windows Registry information can be processed to identify artifacts such as recently accessed documents and attached USB devices.

Multimedia analysis: Images can be examined, EXIF information can be extracted, and supported media can be viewed within the application.

Data carving: Autopsy can use carving capabilities to recover certain deleted files from unallocated space when the underlying data is still available.

9. Mobile and Android Support

Autopsy is primarily known for computer and disk forensics, but it also provides mobile-related capabilities. Its current documentation includes an Android Analyzer ingest module, and the platform can examine supported mobile artifacts.

However, Autopsy should not be confused with a dedicated mobile-acquisition suite. In a smartphone investigation, obtaining a comprehensive extraction may require a separate acquisition tool or an already-created forensic image/backup. Autopsy can then be used to analyze supported data within its case environment.

10. Advantages

- **Open source:** the platform is freely available and its architecture can be inspected and extended.
- **Strong disk and filesystem analysis:** it is built around The Sleuth Kit's forensic capabilities.
- **Modular architecture:** ingest, viewer and reporting modules can be added or extended.
- **Large artifact coverage:** many common operating-system, browser, file and application artifacts can be processed automatically.
- **Parallel processing:** ingest modules can operate in parallel to improve processing efficiency.
- **Integrated workflow:** evidence processing, analysis, tagging, correlation and reporting are available through one case interface.
- **Educational value:** its open architecture makes it particularly useful for learning digital-forensic concepts and workflows.

11. Limitations

- **Not a universal acquisition tool:** Autopsy primarily performs examination and analysis. Obtaining evidence from a locked or damaged device may require specialized acquisition hardware or software.
- **Processing can be resource-intensive:** large disk images may take considerable time and storage to process, especially when many ingest modules are enabled.
- **Artifact support is not unlimited:** new applications, proprietary databases and unusual formats may require additional modules or manual analysis.

- **Automated results require validation:** parsed artifacts can contain interpretation errors or incomplete context, so important findings should be checked against underlying evidence.
- **Mobile acquisition is limited compared with dedicated mobile-forensics suites:** support exists for mobile artifacts, but it does not replace specialized acquisition platforms for every Android or iOS scenario.
- **Technical knowledge is still required:** although the GUI simplifies the workflow, investigators must understand filesystems, timestamps, metadata, hashes, databases and forensic methodology.
- **Deleted-file recovery is not guaranteed:** carving can recover data only when the relevant content has not been overwritten or otherwise destroyed.

12. Reporting Architecture

Autopsy includes an extensible reporting system. Standard reports can include HTML and Excel outputs, while body-file and KML-related reporting are also supported. Report modules can access results generated during the investigation and package selected findings for later review.

The reporting stage is therefore connected to the central case database rather than requiring the investigator to manually copy every finding. Tags, comments, search results and other processed artifacts can be included according to the selected report configuration.

13. Autopsy in a Digital-Forensics Architecture

The complete role of Autopsy can be represented as:

Stage	Function
Evidence acquisition	A forensic image, local drive or logical files are obtained using an appropriate acquisition procedure.
Case creation	The evidence is placed into an Autopsy case.
Data-source processing	The source is parsed using The Sleuth Kit and other supported components.
Ingest processing	Selected modules perform hashing, keyword search, web analysis, file identification, carving and other tasks.
Central results	Module results are stored and organized in the case database.
Investigator analysis	The examiner reviews artifacts, searches, timelines, files and metadata and tags significant evidence.
Reporting	Selected findings are exported into forensic reports.

14. Conclusion

Autopsy is an open-source digital-forensics platform that provides a graphical and modular environment for examining digital evidence. Its architecture combines The Sleuth Kit and other forensic components with a case-management system, ingest-module framework, central results database, analysis interface and reporting system.

Its major strength is its modular approach. Instead of depending on one monolithic analysis mechanism, Autopsy processes evidence through specialized modules for tasks such as filesystem analysis, hashing, keyword search, web-artifact extraction, timeline analysis, carving and other forensic operations. The results are brought together in a common case environment where an investigator can search, correlate, validate and report findings.

Its main limitations are that it does not eliminate the need for proper evidence acquisition or investigator expertise, and its artifact coverage and recovery capabilities are dependent on the underlying evidence and available modules. For mobile investigations in particular, it can analyze supported mobile artifacts but should not be assumed to replace dedicated mobile-acquisition tools.

In short: Autopsy provides the forensic case environment and analysis pipeline; The Sleuth Kit provides important low-level forensic capabilities; ingest modules perform specialized processing; and the investigator interprets and reports the resulting evidence.

15. References

1. Sleuth Kit Labs, Autopsy — official product and platform documentation.
2. Autopsy User Guide — official workflow, case, data-source and ingest-module documentation.
3. Autopsy Features — official documentation of analysis features and supported input/report formats.
4. Autopsy Developer Documentation — official documentation of the modular architecture and extension framework.